

Final Deliverable Report

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1. Data

A. Description of your data set: what is the data about?

The data is about my daily sleep data. It includes date and sleep start time, deep sleep duration, light sleep duration, wake episodes, wake duration, average heart rate, sleep score and my subjective feeling.

B. Motivation: Why did you decide to collect this data?

Sleep is essential for our health and a part of everyone's daily routine, analysing sleep data may give a way to improve health. I have a fitness bracelet, it's easy to get such data.

C. Three data questions you have formulated about the dataset you have collected, providing the following details for each data question

- Is there any hidden pattern in my sleep data?

Why are you interested in this question?

I want to find hidden patterns in my sleep data because sleep is a mystery. Looking at 14 days together, instead of just one by one, can show me the bigger picture of how I really rest. This helps me understand my own body's natural rhythms.

Why do you think physicalization can be well suited to answering this question?

Putting the physicalization models together for two weeks makes it easy to see whether the sleep patterns are similar in the two weeks.

- Will waking up affect the sleep quality and the duration of deep sleep?

Why are you interested in this question?

Waking up may not seem to reduce sleep time much, but sleep interruption is likely to affect sleep quality.

Why do you think physicalization can be well suited to answering this question?

I can easily identify physicalization models with frequent or prolonged awakenings, and then observe whether sleep quality is generally poor in these models.

- Does the objective sleep score match my subjective perception? And in which situations did there exist significant discrepancies?

Why are you interested in this question?

I want to see if the numbers agree with how I really felt. Finding these differences can help me understand what truly makes my sleep feel good or bad.

Why do you think physicalization can be well suited to answering this question?

If the sleep score number and my feeling don't match, the physical model will look wrong or unbalanced.

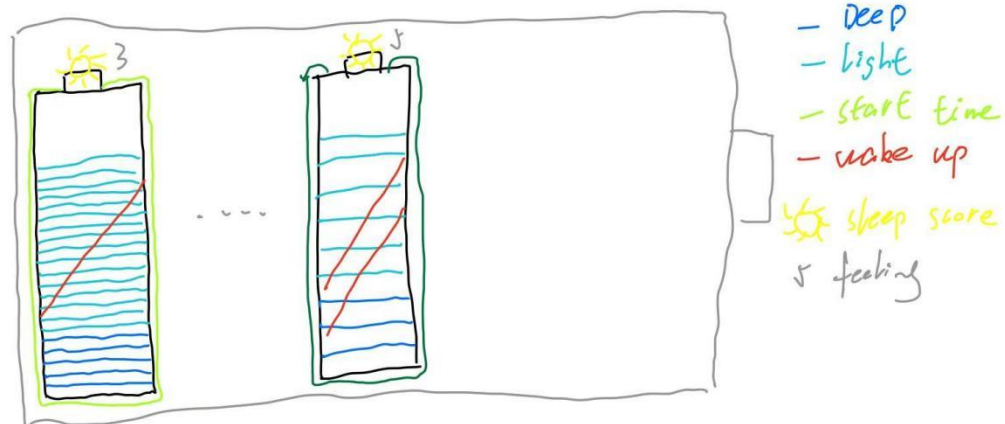
D. Include the dataset

Date and sleep start time Temporal	Deep sleep time Quantitative	Light sleep time Quantitative	Wake episodes Quantitative	Wake duration Quantitative	Average heart rate Quantitative	Sleep score Quantitative	Subjective feeling Ordinal
12/9 00:15	1.8	5.5	1	8	58	82	4
13/9 01:45	1.5	7.2	2	15	60	78	3

14/9 00:30	2.0	6.0	1	5	57	88	5
15/9 23:45	1.9	5.8	0	0	56	89	5
16/9 00:45	1.6	5.9	3	22	51	75	3
17/9 23:50	2.1	5.5	1	10	57	86	4
18/9 23:55	1.7	5.7	2	18	59	80	4
19/9 00:30	1.8	6.0	1	9	58	83	4
20/9 02:00	1.4	7.5	3	25	62	72	2
21/9 23:20	2.2	5.6	0	0	55	92	5
22/9 23:40	2.0	5.7	1	7	56	88	5
23/9 01:30	1.3	5.2	4	35	63	65	2
24/9 23:50	2.0	6.1	2	16	58	84	4
25/9 23:45	1.9	5.9	1	8	57	87	5
26/9 00:20	1.7	6.2	1	12	59	81	4

2. Ideation

A. Battery



- The main body is made of acrylic.
- Deep and Light Sleep are shown by wrapping dark blue and light blue thread around it. The area each color covers matches their proportion of sleep.

- Heart Rate is shown by the density of the threading. The higher heart rate, the tighter the threads are wound.
- Wake-Ups are represented by diagonal red lines. Each line is one wake-up, and its thickness shows how long I was awake.
- Sleep Start Time is indicated by the color of the border around the battery.
- Sleep Score is shown by a light bulb on top. The brighter it is, the higher score.
- Subjective Feeling is written as a number next to the battery.

3. Design critique

A. Battery

It might be tricky to build. Wrapping the thread tightly for heart rate and making the slanted red lines could take some patience! But the result would be awesome.

Reflections on the feedback and which design you want to proceed with. Also include any modifications you would like to do to your chosen design that is based on the feedback you received.

Sleep for humans is like charging, and the first design fits the theme best.

Additionally, the first design makes good use of all the data and is easier to observe and compare. I will continue using the first design.

4. Final design

A. Description and photos of your final physicalization design

The final physicalization design follows the ideation of sleep battery but with some changes. Due to time and material constraints, I only physicalized the data collected for one week, specifically from September 20th to September 26th. A large, battery-shaped wooden board contains seven smaller, battery-shaped wooden boards. Each small battery represents a day's worth of sleep data, and the seven small batteries represent a week's worth of sleep data. Each small battery is wrapped with black and white wires to represent deep sleep and light sleep, respectively. A varying number of small red pins are inserted into each small battery, representing the number of times I woke up during the day. The horizontal height of each small battery varies depending on the overall sleep quality that night. In addition, each small battery has a clock-like dial drawn above it, representing the time when sleep begins that night.

B. Explanation of the visual mapping of data attributes

On each small battery, the area occupied by the black and white wires represents the proportion of total sleep time spent in deep sleep or light sleep, respectively. The density of the thread winding is determined based on the average heart rate during sleep of that day, and is divided into three levels. Specifically, when the average heart rate is 55-57, the spacing between the two strands should be 12 mm; 9 mm when the heart rate is 58-60; and 6 mm when the heart rate is 61-63. The number of red pins on each small battery represents the number of times it wakes up. To keep the physical representation simple, there is no corresponding mapping for the wake-up duration. Each small battery is raised to a different height, based on the overall sleep score given by the bracelet, resulting in four different levels, with a 2mm difference between adjacent levels. There is no mapping for subjective score. The sleep start times are represented by a "clock", with the earliest start time being 11 pm on the left and the latest start time being 2 am on the right to make the difference more obvious.

Process: First, the wooden board is laser-cut into the shape of a battery. Drill a corresponding number of small holes in each small battery to place red pins. Wrap the black and white strings around the small battery according to the corresponding data, and secure them with glue on the back. Based on sleep scores, place a different number of small wooden sticks under each battery to elevate it to different heights, and then secure it with glue. Place the small battery in the corresponding position and insert the red pins. Finally, use a pen to draw a clock above the small battery to indicate the start time of sleep.

D. Reflections on the final design

Through this project, I realized some main differences between physicalization and digital visualization. Physicalizations are things you can touch and interact with in the real world, these make abstract numbers feel more real and memorable because they engage senses beyond just sight. However, my physicalization does not contain any interactive features, it might be improved in the future. In my physicalization, I successfully mapped a large number of data into different representations. Initially, I mainly used those basic methods to represent, like adding a lot of different colors. This may make the physicalization look fancy, but it also makes people confused and cannot distinguish different dimensions that need to be shown. Instead, I added feature like density and volume to the physicalization. I found that although people still need to realize a lot of data visualizations, it's much easier for them to memorize and understand it because of different methods used to represent. For the questions I asked before physicalize the dataset, my work can easily answer that. Due to different representations, it's easy to find the data dimension you concerned and compare it with another dimension. For example, I can find those nights I woke up frequently through checking the number of red pin, and on those batteries, the density of wired strings are high, which means the average heart rate is high when I wake up many time, also, the height (volume) of those batteries is normally lower, this means the sleep quality is bad. There are also some shortcomings of my work. After completing the work, I realized that wiring strings might not be a good idea. Although I fixed the position of each string, it's still quite loose. After a long distance of transportation, the strings can become quite messy. This makes it hard to distinguish batteries through density, and it also takes time to place those strings into right places. Also, the material chosen is not very good. A thick wooden board is hard to drill so it's not easy to fix pins on it. Using light for the sleeping quality might be a good idea but it's very hard to control the brightness so I have to give up, but this also gives me opportunity to find other good representation method. Last but not least, the physicalization process is quite interesting! From brainstorming, to representation learning, to material searching, and to final operation and assembling, this process gives me a sense of achievement. I enjoy it very much!